Project: **Tenant-ive Plans**

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1. We will be studying how certain tenant features can affect the probability of that tenant moving out of a residential rental community in a given week. High tenant turnover at an apartment complex can be an expensive cost. Each time a tenant leaves the property owner loses rent revenue, must pay turnover costs, and risks vacancy. Gated Rentals currently uses a rough assumption (a fixed commute-distance rule) as one factor in determining their tenant turnover rate. They are requesting a data-backed prediction model that can help estimate how long a tenant will stay and roughly predict if a tenant will move out in a given week. Our model hopes to consider how various tenant features such as average tenant income, average tenant family size, average commute to work, and more can increase or decrease the probability of moving out.

2. Gated Rentals has provided us an approved dataset to use. It is an anonymized extract from their property management system covering two properties, Willis Lakes and Crosby Heights. The data is comprised of 455 households across the two properties with 7,566 billing cycles between them. Some features from the data include household incomes, household size, move-in/move-out dates, commute information, and more.

3. We'll build a feed-forward neural network that takes a tenant's features for a given week and outputs the probability that tenant leaves during that specific week (a discrete-time departure (hazard) model). We'll compare it against two baselines provided by the client. First is a purely descriptive survival curve with no individual features at all (a Kaplan-Meier estimate) which ignores tenant differences entirely. Second is a logistic regression model trained on the exact same inputs as our neural network, which represents a simpler, classical model with the same information. This lets us show specifically what value, if any, the neural network's added flexibility contributes beyond a simple model

4. We'll evaluate predictions at fixed horizons of 26 and 52 weeks using calibration and a Brier score to compare our neural model against both baselines. We will also run an ablation and train the model with and without commute-related features to determine how much influential commute burden can be to the predictions. Lastly, we will rerun our key results on the highest-confidence subset of the data to confirm any patterns and remove any noise. Ultimate success will be achieved when the model closely matches historical data of tenant tenure given by the client.

5. The project at its core is a classification predictive model: "Is the tenant moving out or not?". We are focusing on tenant-level retention. The dataset is already cleaned, structured, and documented by the client. A feed-forward neural network of this size trains in minutes on a laptop or a free cloud notebook, so no GPU cluster is required. With three team members, we can divide the work along the course's own milestones: data preparation and baseline models before the midterm report, the neural model, ablation, and fairness checks before the final report.

Data Set: https://github.com/pvo18/Team-Carrier-Pigeon/blob/main/Gated_Rentals_Dataset.zip

References:

- Ezenwaka, Chiamaka. (2025). Integrating AI-Powered Business Intelligence Dashboards to Forecast Commercial Property Trends and Tenant Retention Metrics. International Journal of Computer Applications Technology and Research. 47-62. 10.7753/IJCATR1408.1006. https://www.researchgate.net/publication/394342000_Integrating_AI-Powered_Business_Intelligence_Dashboards_to_Forecast_Commercial_Property_Trends_and_Tenant_Retention_Metrics

- Harrison, Thomas. (2026). Artificial Intelligence in Property Management: Improving Leasing, Tenant Retention, and Operational Efficiency in Multifamily. 9. https://www.researchgate.net/publication/414414920_Artificial_Intelligence_in_Property_Management_Improving_Leasing_Tenant_Retention_and_Operational_Efficiency_in_Multifamily